"Deep Signal" - Stochastic Oscillators and Volume

Applying a low-frequency strategy to a medium-frequency algorithm

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Algorithm-Based Low-Frequency Trading Using a Stochastic Oscillator, Williams%R, and Trading Volume for the S&P 500

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Abstract: Recent research in algorithmic trading has primarily focused on ultra-high-frequency strategies and index estimation. In response to the need for a low-frequency, real-world trading model, we developed an enhanced algorithm that builds on existing models with high hit ratios and low maximum drawdowns. We utilized established price indicators, including the stochastic oscillator and Williams %R, while introducing a volume factor to improve the model's robustness and performance. The refined algorithm achieved superior returns while maintaining its high hit ratio and low maximum drawdown. Specifically, we leveraged 2X and 3X signals, incorporating volume data, the 52-week average, standard deviation, and other variables. The dataset comprised SPY ETF price and volume data spanning from 2010 to 2023, over 13 years. Our enhanced algorithmic model outperformed both the benchmark and previous iterations, achieving a hit rate of over 90%, a

Definitions

- **Williams%R:**

$$\%R = \frac{\text{Close} - \text{Highest High}_n}{\text{Highest High}_n - \text{Lowest Low}_n} \times 100 \tag{1}$$

- **Stochastic Oscillator Components:**

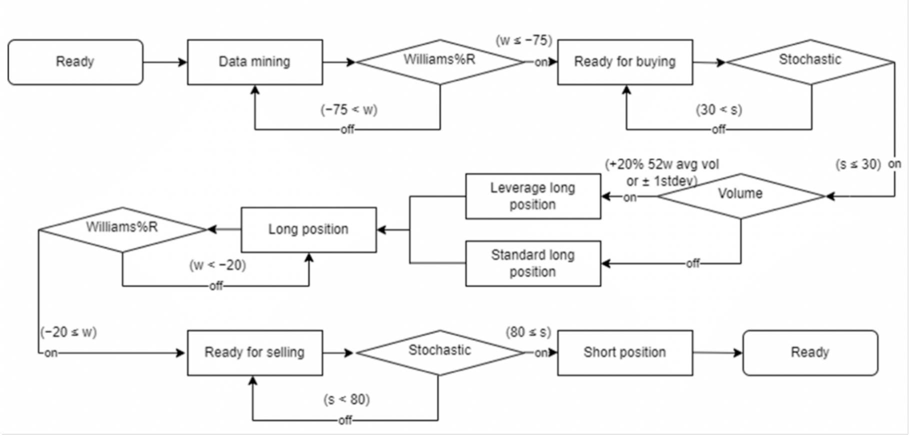
$$\%K = \frac{\text{Close} - \text{Lowest Low}_n}{\text{Highest High}_n - \text{Lowest Low}_n} \times 100 \tag{2}$$

$$\%D = \frac{1}{m} \sum_{i=0}^{m-1} \%K_{t-i} \tag{3}$$

The Intuition

- %R and %D very high (near-0 in %R case, near-100 for %K) implies the asset is trading near its (recent) maximum, and so we tend to call it overbought. Same vice versa.
- Volume very high often leads to strong trends and is correlated with high price movements.
- "during a high trading volume period, markets are more likely to be dominated by overconfident traders"
- The logic is: wait for the signals $\rightarrow$ Size trade based on volume $\rightarrow$ Wait for the signal to disappear to exit.

The Original Algorithm



Changing Frequency

- We have a problem! 2010-2023, this algorithm traded 10 times!
- New universe: Top X SNP 500 Constituents
- Let X $= 40$ (arbitrarily)

Results

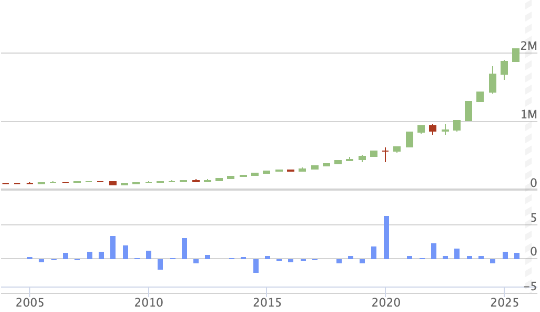


Figure: Results

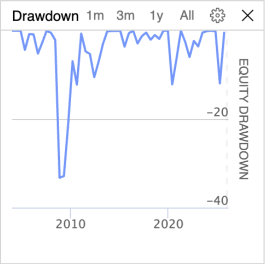


Figure: Drawdown

Summary of Results

- Sharpe Ratio: 0.545, Drawdown: 47.4%
- Compounding Annual Return: 14.9%
- We can improve Sharpe and CAR by increasing the total exposure of the strategy, but this comes at a steep cost to Drawdown.
- What's wrong?

Why the poor performance?

- "Grid Searching"
- 2008
- Strength of the Signal?
- Higher volatility of Constituents

Thank you for listening

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